

№8

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1

$$\begin{cases} \frac{dM_1}{d\theta} = M_1 - \frac{b}{c_1} M_1 M_2 - \frac{a_1}{c_1} M_1^2, \\ \frac{dM_2}{d\theta} = \frac{c_2}{c_1} M_2 - \frac{b}{c_1} M_1 M_2 - \frac{a_1}{c_1} M_2^2, \end{cases}$$

$$a_1 = \frac{p_{cr}}{\tau_1^2 \tilde{p}_1^2 Nq}, \quad a_2 = \frac{p_{cr}}{\tau_2^2 \tilde{p}_2^2 Nq}, \quad b = \frac{p_{cr}}{\tau_1^2 \tilde{p}_1^2 \tau_2^2 \tilde{p}_2^2 Nq}, \quad c_1 = \frac{p_{cr} - \tilde{p}_1}{\tau_1 \tilde{p}_1}, \quad c_2 = \frac{p_{cr} - \tilde{p}_1}{\tau_2 \tilde{p}_2}$$

$$t = c_1 \theta.$$

2

$$\begin{cases} \frac{dM_1}{d\theta} = M_1 - \left(\frac{b}{c_1} + 0.00015\right)M_1M_2 - \frac{a_1}{c_1}M_1^2, \\ \frac{dM_2}{d\theta} = \frac{c_2}{c_1}M_2 - \frac{b}{c_1}M_1M_2 - \frac{a_1}{c_1}M_2^2, \end{cases}$$

$$\begin{aligned}M_0^1 &= 7.3, M_0^2 = 8.3, \\p_{cr} &= 42, N = 88, q = 1 \\ \tau_1 &= 28, \tau_2 = 25, \\ \tilde{p}_1 &= 13, \tilde{p}_2 = 10\end{aligned}$$

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- N – .
- τ –
- p –
- $\tilde{p}$ – , .
- q –
- $\theta = \frac{t}{c_1}$ –

1.

1.

2.

2.

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2

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2



Julia
Plots.jl

DifferentialEquations.jl

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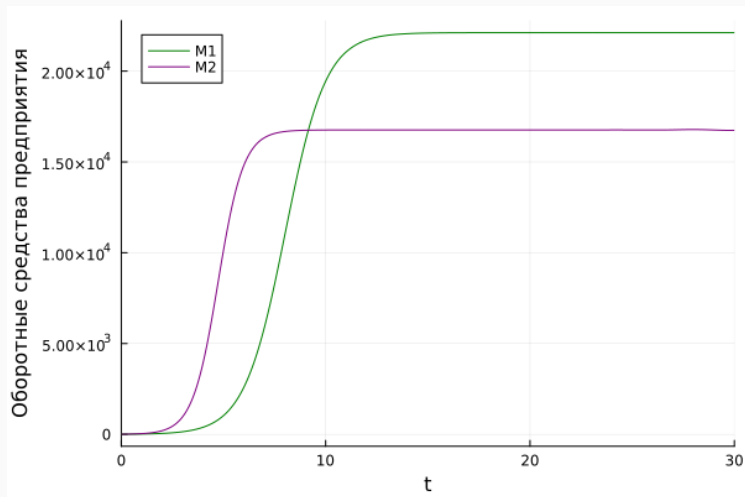
```
p_cr = 42 #  
tau1 = 28 #  
p1 = 13 #  
tau2 = 25 #  
p2 = 10 #  
N = 88 #  
q = 1; #
```

		1
	1	
		2
	2	

```
a1 = p_cr/(tau1^2*p1^2*N*q);  
a2 = p_cr/(tau2^2*p2^2*N*q);  
b = p_cr/(tau1^2*tau2^2*p1^2*p2^2*N*q);  
c1 = (p_cr-p1)/(tau1*p1);  
c2 = (p_cr-p2)/(tau2*p2);  
  
u0 = [7.3, 8.3] # M1 M2  
p = [a1, a2, b, c1, c2]  
tspan = (0.0, 30.0) #
```

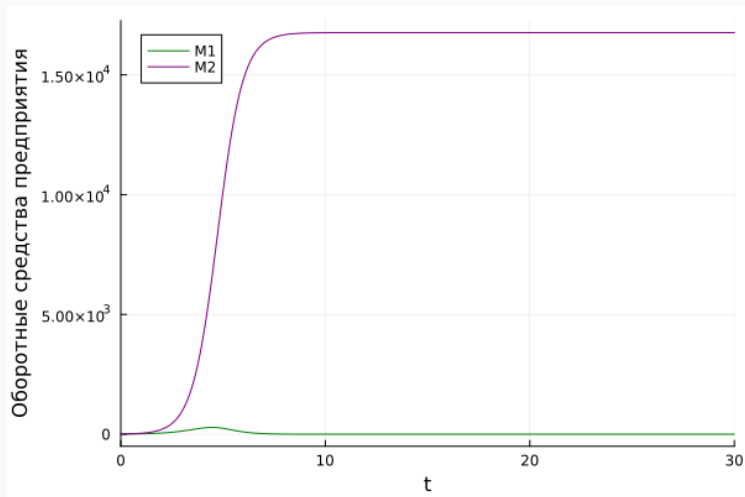
```
function f(u, p, t)
    M1, M2 = u
    a1, a2, b, c1, c2 = p
    M1 = M1 - (a1/c1)*M1^2 - (b/c1)*M1*M2
    M2 = (c2/c1)*M2 - (a2/c1)*M2^2 - (b/c1)*M1*M2
    return [M1, M2]
end
```

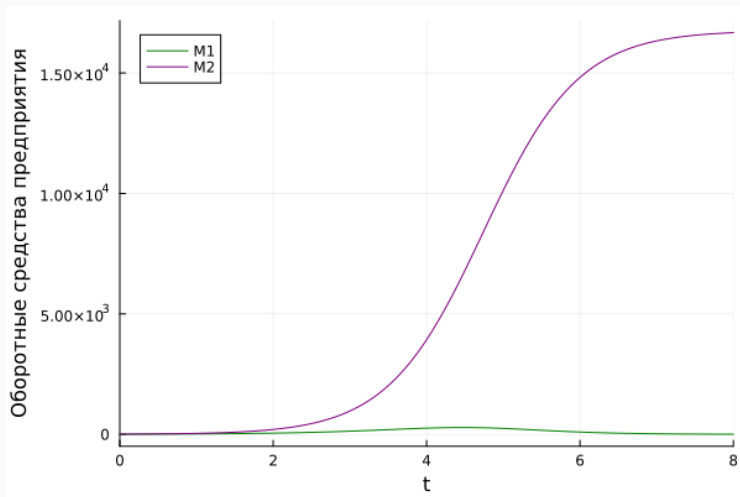
```
prob = ODEProblem(f, u0, tspan, p)
sol = solve(prob, Tsit5(), saveat = 0.01)
plot(sol, yaxis = "          ", label = ["M1" "M2"],
      c = ["green" "purple"])
```

```
function f(u, p, t)
    M1, M2 = u
    a1, a2, b, c1, c2 = p
    M1 = M1 - (a1/c1)*M1^2 - (b/c1+0.00015)*M1*M2
    M2 = (c2/c1)*M2 - (a2/c1)*M2^2 - (b/c1)*M1*M2
    return [M1, M2]
end
```

```
prob = ODEProblem(f, u0, tspan, p)
sol = solve(prob, Tsit5(), saveat = 0.01)
plot(sol, yaxis = "          ", label = ["M1" "M2"],
      c = ["green" "purple"])
```



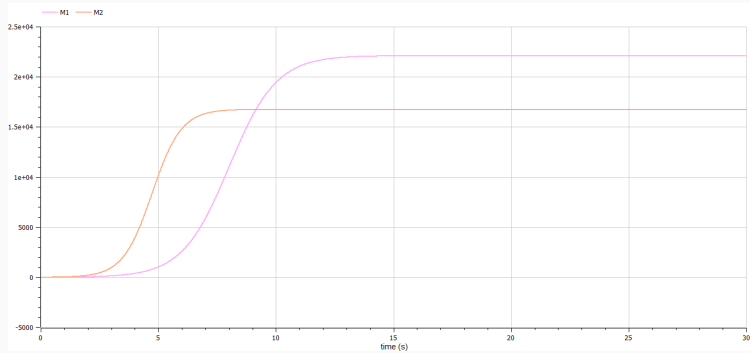


```
parameter Real p_cr = 42;  
parameter Real tau1 = 28;  
parameter Real p1 = 13;  
parameter Real tau2 = 25;  
parameter Real p2 = 10;  
parameter Real N = 88;  
parameter Real q = 1;  
parameter Real a1 = p_cr/(tau1^2*p1^2*N*q);  
parameter Real a2 = p_cr/(tau2^2*p2^2*N*q);  
parameter Real b = p_cr/(tau1^2*tau2^2*p1^2*p2^2*N*q);  
parameter Real c1 = (p_cr-p1)/(tau1*p1);  
parameter Real c2 = (p_cr-p2)/(tau2*p2);  
Real M1(start=7.3);  
Real M2(start=8.3);
```

equation

$$\text{der}(M1) = M1 - (a1/c1)*M1^2 - (b/c1)*M1*M2;$$

$$\text{der}(M2) = (c2/c1)*M2 - (a2/c1)*M2^2 - (b/c1)*M1*M2;$$



. 4:

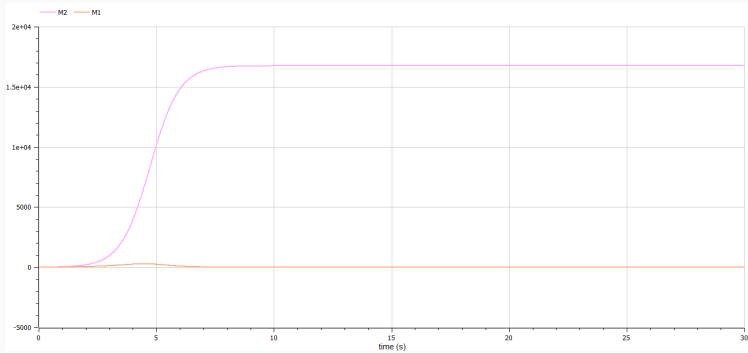
1

2


```
parameter Real p_cr = 42;  
parameter Real tau1 = 28;  
parameter Real p1 = 13;  
parameter Real tau2 = 25;  
parameter Real p2 = 10;  
parameter Real N = 88;  
parameter Real q = 1;  
parameter Real a1 = p_cr/(tau1^2*p1^2*N*q);  
parameter Real a2 = p_cr/(tau2^2*p2^2*N*q);  
parameter Real b = p_cr/(tau1^2*tau2^2*p1^2*p2^2*N*q);  
parameter Real c1 = (p_cr-p1)/(tau1*p1);  
parameter Real c2 = (p_cr-p2)/(tau2*p2);  
Real M1(start=7.3);  
Real M2(start=8.3);
```

equation

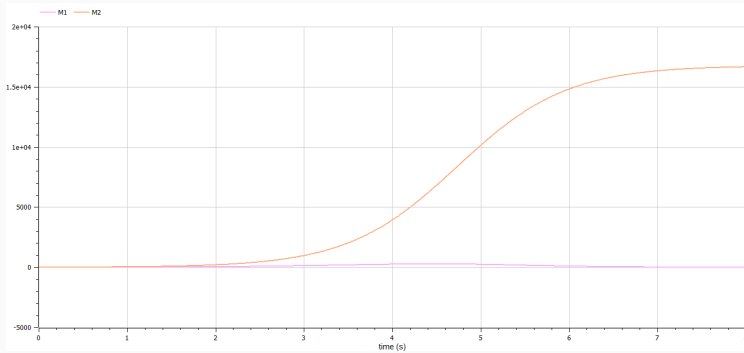
$$\text{der}(M1) = M1 - (a1/c1)*M1^2 - (b/c1+0.00015)*M1*M2;$$
$$\text{der}(M2) = (c2/c1)*M2 - (a2/c1)*M2^2 - (b/c1)*M1*M2;$$



. 5:

1

2



. 6:

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1. . . , 2003. 29-32 .